Project Proposal

Project Title: Detection and Prevention of Malware and Malicious File Injection in RPA/Drone Feeds

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Sponsoring Agency: Bharat Electronics Ltd

Cost: INR 33.748 Lakhs, Duration: 18 Months

Abstract: This proposal outlines a comprehensive solution for detecting and preventing malicious files or malware potentially injected during the ingestion of data (videos/images/target metadata) from Remotely Piloted Aircraft (RPAs)/drones into edge or operational networks. The objective is to ensure the integrity and security of digital feeds before they are disseminated further within the network.

1. Problem Statement

In contemporary mission-critical environments, Remotely Piloted Aircraft (RPAs) and drones serve as vital assets for surveillance, reconnaissance, and real-time intelligence gathering. These platforms transmit a continuous stream of data, including *live video feeds, images, telemetry, and sensor metadata* to edge networks and command systems for immediate analysis and action. However, this creates a highly exposed attack surface, as adversaries can attempt to inject malware through seemingly legitimate data streams. Malicious payloads may be embedded in multimedia files, obfuscated in *encrypted files, ZIP archives*, or hidden within metadata fields such as EXIF or document scripts.

Once injected, such malware can compromise sensitive infrastructure, exfiltrate classified data, or manipulate critical operations. Given the increasing reliance on automated, low-latency decision-making in military and strategic applications, it becomes essential to detect and prevent malware at the ingestion point itself, before it affects operational integrity. The solution must be intelligent, adaptable, and capable of handling evolving threats, all while maintaining real-time performance and compatibility with field-deployed drones.

2. Objectives of this Project

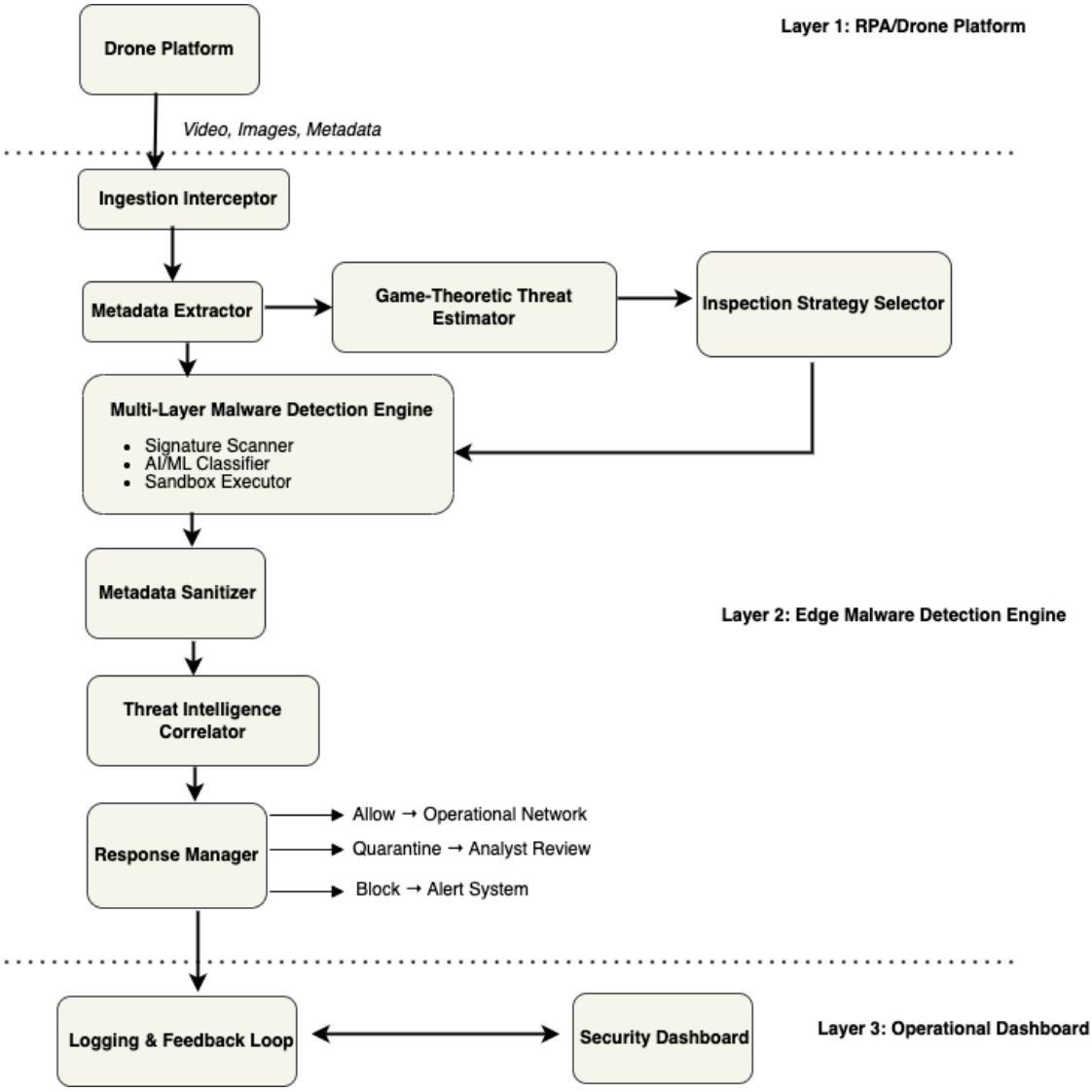
The primary goals of this project are as follows:

1. *Detect and Block Malicious Payloads:* Identify and neutralise malicious payloads (images, videos, telemetry, ZIP/encrypted files, live-streaming data) from RPAs/drones at the point of ingestion, before they enter the edge network.
2. *Device Authentication and Metadata Validation:* Verify the legitimacy of transmitting drones using metadata (timestamps, drone ID, geolocation) to prevent spoofed or unauthorised sources..
3. *Implement Risk-Based Adaptive Inspection:* Use contextual threat scoring to dynamically determine the depth and method of inspection, optimising resource usage based on the origin, type, and behaviour of incoming files.
4. *Integrate AI/ML and Game-Theoretic Defence:* Leverage artificial intelligence and game theory models to adaptively anticipate and counter sophisticated adversarial tactics.
5. *Enable Full Visibility and Auditability:* Ensure traceability, generate security logs, and provide actionable alerts for operators to facilitate investigation and response.

6. *Ensure Real-Time Performance:* Maintain near-zero-latency processing to ensure uninterrupted drone-to-network data transmission and timely threat response.

3. Proposed Architecture

To mitigate malware injection from Remotely Piloted Aircraft (RPA) or drone feeds, we propose a *Multi-Layered Malware Detection and Threat Prevention System* optimized for edge deployment. The system inspects ingested data such as live video, images, telemetry, and encrypted files in real-time using signature-based, AI/ML, and sandbox analysis. It also supports *two-way communication,* ensuring that both incoming data from drones and outgoing control commands from the operations center are validated and secure. This helps maintain integrity and security across the full communication loop. The architecture is adaptive, intelligent, and designed to counter evolving threats with minimal impact on mission-critical operations.



Proposed Architecture

Key Entities / Modules

1. RPA/Drone Data Source

- **Function:** Acts as the origin of data feeds, including video, images, encrypted/ZIP files, telemetry, text, and metadata.
- **Role:** Transmits mission-critical data to the edge/operational network in real time.

2. Ingestion Interceptor

- **Function:** Initial capture, authentication, and verification of incoming data streams.
- **Responsibilities:**
  - Authenticate source devices using metadata (e.g., timestamps, drone ID, geolocation).
  - Validate data format, structure, and integrity.
  - Extract and record metadata: drone ID, mission zone, timestamps, encryption status, etc.
  - Identify ZIP/encrypted file types for deferred analysis.
  - Act as a secure data funnel, preventing uncontrolled input into the system.
  - Support secure uplink communication from control center, enabling:
    - Receipt of quarantine or response commands based on detected threats.
    - Dynamic adjustment of inspection parameters or AV/ML module updates.
    - Live feedback loop for mission-critical interventions.

3. Game-Theoretic Threat Estimator

- **Function:** Intelligent risk evaluator modelling attacker-defender behaviour.
- **Capabilities:**
  - Leverages Stackelberg game theory to simulate adversarial behaviour under asymmetric knowledge.
  - Uses Bayesian game modelling for probabilistic estimation in uncertain threat scenarios.
  - Computes a dynamic **Threat Score (T_S)** based on:
    - Metadata irregularities and anomalies.
    - Source reliability and authentication history.
    - Current operational context (mission sensitivity, location).
  - Advises the Inspection Strategy Selector on optimal analysis path (lightweight vs. deep scan).

4. Multi-Layer Malware Detection Engine

A progressive analysis stack is activated based on Threat Score:

a. Signature-Based Anti-Virus (AV) Scanner

- Detects known malware using licensed signature databases.
- Quick filtering for known threats.

b. AI/ML Malware Classifier

- Predicts maliciousness using file-based features (entropy, file size, format, structure).
- Technologies: Trained machine learning models (Support Vector Machine (SVM), Random Forest (RF), eXtreme Gradient Boosting (XGBoost), or Deep Learning (DL) models).
- Handles encrypted traffic through heuristic pre-analysis.

c. Behavioral Sandbox Analyzer

- Executes suspicious files in a virtual container to monitor behaviour.
- Monitors include:
  - File system activities.
  - Registry changes.
  - API calls and network communication attempts.
- Supports sandboxing ZIP/encrypted files after decryption.

5. Metadata Sanitisation Module

- **Function:** Prevents attacks via embedded metadata.
- **Actions:**
  - Scrubs potentially harmful fields from EXIF, PDF, DOC, and multimedia files.
  - Removes tracking tags, hidden scripts, and malformed metadata entries.
  - Normalises metadata across video/image files to avoid fingerprinting or malware delivery.

6. Threat Intelligence Co-relator

- **Function:** Verifies suspicious artifacts with external and internal threat intelligence.
- **Data Sources:** Validate and confirm suspicious file behaviour with known indicators of compromise (IOC).
- **Purpose:**
  - Corroborate suspicious findings.
  - Augment internal knowledge base with real-world IOCs.

- Guide automated decision-making with contextual threat understanding.

7. Response and Quarantine Manager

- **Function:** Acts based on the final threat classification.
- **Decisions:**
  - Allow: Forward clean files/data for further processing.
  - Isolate: Hold flagged files for manual inspection.
  - Block & Alert: Quarantine confirmed threats and notify operators immediately.

8. Security Feedback Loop

- **Function:** Closes the learning loop for adaptive defence.
- **Purpose:**
  - Update ML models with true/false positive outcomes.
  - Refine game-theoretic threat models based on evolving adversary strategies.
  - Improve real-time classification over time.

9. Security Dashboard & Reporting System

- **Function:** Offers a centralised visibility and analytics interface.
- **Features:**
  - Real-time monitoring of threat scores, system status, and file disposition.
  - Incident logs, forensic reports, and traceability for audit readiness.
  - Compliance metrics and SLA violation reporting.

Key Components and Their Responsibilities

Entity/Module	Responsibility
Drone/RPA Data Source	Sends mission-critical data (images, video, telemetry, encrypted files metadata, live-streaming) to the edge/operational network in real time.
Ingestion Interceptor	Captures, authenticates, and pre-validates incoming data; extracts an records metadata; flags encrypted/ZIP content for deep inspection.
Threat Estimator (Game-Theoretic)	Assigns a dynamic threat score using Stackelberg and Bayesian models based on metadata anomalies, source trust, and operational context.

Multi-Layer Malware Detection Engine	Performs layered malware inspection including: 1) Signature-based AV scanning, 2) AI/ML-based classification, and 3) behavioural sandbox analysis.
Metadata Sanitizer	Cleans embedded metadata to remove hidden threats.
Threat Intelligence Correlator Response & Quarantine Manager	Validates behavioural signatures against known Indicators of compromise (IOCs). Allows, blocks, or isolates files based on risk.
Security Dashboard & Feedback Loop	Provides real-time visual threat analytics, logs incidents, and feeds detection outcomes back into ML/game-theoretic models to improve future accuracy.

Scalability and Throughput Requirements

- The system is designed to handle a throughput of up to 1 Gbps, supporting 300 – 400 concurrent drone/RPA video and telemetry feeds.
- The ingestion pipeline, threat estimation, and malware detection modules are optimized for parallel processing and low-latency analysis across multiple input streams.
- Real-time performance is prioritized to ensure mission-critical responsiveness without bottlenecks.

4. Workflow

The proposed malware detection system for drone and RPA (Remotely Piloted Aircraft) data streams begins by intercepting all incoming files, such as images, videos, telemetry, encrypted archives, and metadata, at the edge through a secure ingestion interceptor. This module extracts, parses, and validates data formats, authenticates the source device based on metadata, and records key attributes such as drone ID, timestamp, mission zone, and geolocation.

Once the data is collected, a game-theoretic threat estimator module applies a Stackelberg adversarial game framework to assess the risk level of each file. This model treats the defender (malware detection system) as the leader who proactively chooses the intensity of inspection, anticipating how an attacker might respond with evasive malware strategies. Based on source reliability, file type (including ZIP or encrypted content), operational context, and content features, the module computes a threat score (T_S) and dynamically adjusts the inspection depth, ranging from lightweight signature scans to full sandbox execution.

Files are then routed to a multi-layer malware detection engine according to their computed threat scores. All files first undergo a signature-based antivirus scan using licensed detection software. Files with medium or high threat

scores are passed through an AI/ML-based classifier trained on behavioural and statistical features. High-risk or obfuscated files are executed within a secure sandbox infrastructure, which monitors real-time behaviours such as process spawning, file system changes, registry edits, and outbound network activity. Additionally, live video feeds are subjected to real-time threat pattern analysis to detect embedded or streamed threats. In parallel, a metadata sanitisation engine inspects and cleans embedded metadata, such as EXIF fields in images, document headers, or hidden scripts, to remove covert channels and tracking payloads. All extracted behavioural indicators and metadata artifacts are correlated with external threat intelligence sources (IOC feeds) to validate and enhance detection fidelity.

Based on cumulative analyses, the response and quarantine manager decides on final actions: clean files are forwarded to the operational systems, while suspicious or malicious content is quarantined, blocked, or flagged for analyst review. If a threat is confirmed, the associated drone feed is marked as compromised, and alerts are sent to the Security Operations Center (SOC) or Security Information and Event Management (SIEM) platforms. All threat assessments, inspection outcomes, and attacker behaviours are logged and looped back into the game-theoretic estimator and ML models to improve future decision-making. The system also includes a real-time dashboard and reporting interface, offering comprehensive visibility into threat levels, malware detection events, and overall system health. It supports periodic analytics reports, incident timelines, and license auditing of the signature-based software, ensuring compliance and audit readiness. This holistic, adaptive architecture enables proactive defence against sophisticated malware threats while sustaining mission-critical performance in edge-based drone ecosystems. The data flow is summarised as follows:

1. **Data Ingestion**
   Drone data (images, video, text, telemetry, encrypted/ZIP files) is received and buffered by the **ingestion interceptor**, which also authenticates the device.

2. **Metadata Extraction**
   Extracts and validates metadata (timestamp, drone ID, location, file type). Flags encrypted and compressed files for deeper checks.

3. **Threat Scoring**
   A **game-theoretic model** assigns a Threat Score (T_S) based on source trust, context, and file traits.
   Inspection levels:
   - *Low*: Signature scan
   - *Medium*: Signature + AI/ML
   - *High*: Sandbox execution

4. **Malware Detection**
   - **Signature Scan**: Licensed AV engine
   - **AI/ML Classifier**: Based on file features
   - **Sandboxing**: Safe execution to observe behaviour
   - **Live Video Analysis**: Scans for threats in real-time feeds

5. **Metadata Cleaning**

   Sanitises embedded data (e.g., EXIF, scripts) to eliminate hidden threats.

6. **Threat Intelligence Matching**

   Cross-checks behaviour and IOCs with internal and external threat feeds.

7. **Response Actions**
   - *Clean*: Forward to system
   - *Suspicious*: Quarantine
   - *Malicious*: Block + alert

8. **Logging & Feedback**

   Logs all activity. Updates ML models and threat scoring with detection outcomes.

9. **Dashboard Monitoring**

   Real-time dashboard shows alerts, analytics, logs, and system status for operators.

5. Simulated Drone Feed Data (Video, Images, Metadata)

1. **Open-Source Drone Datasets**: Utilise publicly available datasets such as:
   - **DroneDeploy**,
   - **UAV (Kaggle)**
   - **UAV123**,
   - **VisDrone**

These contain aerial video/image data along with partial telemetry.

2. **Synthetic Metadata Generation**: Use custom scripts or drone simulators to generate metadata, including:
   - Drone ID, GPS location, timestamps, altitude, mission zone

This helps emulate diverse operational scenarios for robust testing.

6. Core Capabilities

The proposed Multi-Layered Malware Detection and Threat Prevention System for edge-based drone and RPA data ingestion offers several important advantages, enabling robust, intelligent, and adaptive cybersecurity at the operational edge:

- **Proactive Threat Blocking**: Prevents malware from entering the edge network at the earliest ingestion point.
- **Adaptive Inspection**: Optimizes resource usage using risk-based strategies driven by game-theoretic modelling.
- **Multi-Layered Detection**: Combines traditional, AI-based, and behaviour-based analysis for high accuracy.
- **Evolving Defence**: Feedback loop allows continuous learning from past incidents and adversarial tactics.
- **Operational Continuity**: Minimal disruption to data flow ensures real-time performance for mission-critical applications.
- **Comprehensive Visibility**: Centralised dashboard supports decision-making, incident response, and reporting.

7. Deliverables

Phase	Deliverables
Phase 0	SRS Document
Phase 1 (Months 1-6)	Requirements Specification, Architecture Blueprint, Initial Module Designs (Interceptor, Scanner)
Phase 2 (Months 7-12)	Core Modules Developed (Sandbox, AI/ML Classifier), Threat Estimator, Metadata Sanitiser, Basic Dashboard
Phase 3 (Months 13-18)	Pilot Deployment Report, System Integration (SIEM), Automated Response Features, Final Dashboard, Documentation, Training, Handover Report

8. Project Timeline - Implementation Phases (18 Months)

Phase 1: Planning & Core Development (Months 1–6)

- Requirements gathering & threat modelling
- System architecture and design
- Develop an ingestion interceptor and a signature-based malware scanner
- Initial development of threat estimator (game-theoretic model)

- Start metadata sanitisation module

Outcome: Architecture Blueprint, Initial Module Designs (Interceptor, Scanner)

Phase 2: Advanced Module Development & Integration (Months 7–12)

- Develop an AI/ML malware classifier and sandbox analyser
- Integrate threat intelligence correlation and response/quarantine manager
- Build initial dashboard and logging systems
- Internal lab testing and iterative improvements

Outcome: Core Modules Developed (Sandbox, AI/ML Classifier), Threat Estimator, Metadata Sanitiser, Basic Dashboard

Phase 3: Pilot Deployment, Optimisation & Handover (Months 13–18)

- Deploy the pilot system on selected edge nodes
- Conduct live testing with real drone/RPA data (including encrypted traffic and zip files)
- Integrate with SIEM and implement automated response workflows
- Optimise ML models and game-theoretic threat estimator with the feedback loop
- Finalise dashboard features and reporting
- Complete documentation, training, and project handover

Outcome: Pilot Deployment Report, System Integration (SIEM), Automated Response Features, Final Dashboard, Documentation, Training, Handover Report

Milestones & Timeline

Month Range	Milestone	Output
M0	1. Project Kickoff	SRS Document
M1–M6	2. Core Module Development	Architecture, Interceptor, Scanner Prototype
M7–M12	3. Advanced Development & Integration	AI/ML Classifier, Sandbox, Threat Estimator, Basic Dashboard

M12–M18	4. Pilot Deployment & Optimisation	Live Pilot Report, SIEM Integration, Automated Response, Final Documentation, Training

8. References

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3. A. L. Buczak and E. Guven, "A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection," in IEEE Communications Surveys & Tutorials, vol. 18, no. 2, pp. 1153-1176, Secondquarter 2016, doi: 10.1109/COMST.2015.2494502.
4. Kamhoua, C.A., Kiekintveld, C.D., Fang, F. and Zhu, Q. eds., 2021. Game theory and machine learning for cyber security. John Wiley & Sons.
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Annexure - I

Details Expenditure Break-up for Proposed Solution

a) **Development Cost**	**Rs. 8 L**
b) Project Consultancy Charges	**Rs. 14 L**
c) Institute Overhead (30% of Total above rows) (30% of 22 L)	**Rs. 6.6 L**
GST (18% of total) (18% of 28.6 L)	**Rs. 5.148 L**
Total	**Rs. 33.748 L**

Project Payment Schedule

Stage	Percentage	Amount (INR)	Remarks
Milestone 0	20%	Rs. 6.7496 L	On delivery of the SRS Document
Milestone 1	20%	Rs. 6.7496 L	On completion of Phase 1: Architecture & Core Modules
Milestone 2	30%	Rs. 10.1244 L	On completion of Pilot Deployment & Testing
Milestone 3	30%	Rs. 10.1244 L	On full system deployment, documentation & handover